Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Software Requirements Specification

Reference	TBD **[TBD-1]**
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1.0 Purpose and Scope

This document states what the System's software must do. Each requirement is one sentence with "shall", traced to where it comes from, and given a verification method [2, Sec. 1]. Together, the requirements cover everything the earlier documents ask of the software: every control and every must-work and must-not-work function of the hazard analysis, every scenario, capability and objective of the ConOps, and NASA's rules for safety-critical software.

It covers the three software items of the Software Classification: the controller firmware, the onboard computer software and the ground software, all Class B and safety-critical [4, Secs. 3.0 and 4.0]. Bench test tools and firmware build tools follow the rules of their own class and are not covered.

How to read it. Section 3.0 introduces the software items and the table columns. Sections 4.0 to 6.0 give the requirements of each item, and Section 7.0 those that apply to all three. Appendix A shows, item by item, that nothing the earlier documents ask for is left out.

2.0 Documents

[1] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[2] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 5.09 "SRS – Software Requirements Specification".
[3] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Classification*, Issue 1, Revision 1, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 0, Draft.
[6] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
[7] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Architecture*, Issue 1, Revision 0, Draft.

3.0 Overview

3.1 Software Items

A software item is one program that is built, tested and released as a unit [2, Sec. 3.2]. Requirements are organised by software item; the System has three:

Controller firmware Runs on the controller: acts on commands, reports its state, and returns to working firmware on its own.

Onboard computer software Runs on the onboard computer: checks commands and firmware, stores reports, watches the controller, and restores it.

Ground software Runs in the mission operations center: prepares and checks commands and firmware, and keeps every report.

They meet over the radio link, the two data-link connections, the reprogramming link and the operator interface [3, Sec. 3.3].

The ConOps says which item does most functions. For the few it leaves open, this document assigns one:

- the onboard computer and the controller both check the arming command before a dangerous command, so that one fault cannot let it through;
- the controller answers every command and tests itself;
- the onboard computer holds scheduled commands until they run, refuses new firmware during a trial, numbers reports, compares the onboard clock with ground time, and clears the controller's kept data when it restores it.

3.2 Table Columns

- **ID:** REQ-001 onward, in one sequence. An ID is never reused.
- **Requirement:** one claim, in one sentence.
- **Source:** where the requirement comes from. "H-1 control 3" is control 3 of hazard report H-1, and "H-4 cause e" a cause of that report [5, Sec. 6.0]. A hazard named alone, such as "H-14", is a moderate or negligible hazard, which has no report [5, Sec. 4.0]. Scenarios, capabilities and objectives (O) are those of the ConOps [3, Secs. 3.1, 3.5 and 6.0]. SWE-n is rule n of NPR 7150.2D [1, App. C, pp. 56–78]. A shared cause is a finding of the hazard analysis [5, Sec. 5.2]. "Architecture 5.1" is a design decision of the architecture [7, Sec. 5.1].
- **Verified by** [6, Sec. 5.3.1.2.1, p. 92]: TEST, using the product to obtain detailed data; DEMONSTRATION, using it to show the requirement is met, without detailed data; INSPECTION, examining the product or its records; ANALYSIS, modelling or calculation.

4.0 Controller Firmware Requirements

ID	Requirement	Source	Verified by
REQ-001	The controller shall always be in exactly one of the modes OPERATE, TRIAL, FALLBACK and RECOVERY.	Mode table; SWE-134 b	TEST
REQ-002	The functions available in each controller mode shall be those of the mode table of the ConOps.	Mode table; Scenario-425	TEST
REQ-003	The controller shall change mode only along the transitions the ConOps describes.	Mode table; SWE-134 b	TEST
REQ-004	At every start, the controller firmware shall set itself to a known safe state before acting on any command.	SWE-134 a	TEST
REQ-005	The controller firmware shall check every firmware image for damage before running it, and shall not run a damaged image.	H-1 control 2; H-7 control 3; Scenario-610; O-6	TEST

ID	Requirement	Source	Verified by
REQ-006	The controller firmware shall check that a firmware image comes from the operator before running it.	H-2 control 2	TEST
REQ-007	An update shall never write over the confirmed firmware, the running firmware or the last-resort firmware.	H-1 control 1; Scenario-600; O-6	TEST
REQ-008	The controller shall keep a last-resort firmware that no update replaces.	H-1 control 4; Scenario-460	TEST
REQ-009	The last-resort firmware shall share no start-up code, kept data or other code with the main firmware.	H-1 cause g; H-8 cause g; Shared cause: image code and data	INSPECTION
REQ-010	The last-resort firmware shall be able to receive new firmware.	H-3 control 3; Mode table	TEST
REQ-011	New firmware shall run a trial of set length [**TBD-2**] before it is confirmed.	H-3 control 1; Scenario-100; Capability-4	TEST
REQ-012	New firmware shall be judged good only when the operator confirms it, after the controller has reported that it passed its trial tests.	H-3 control 2; Scenario-100	TEST
REQ-013	New firmware not judged good within the trial time shall be given up, and the controller shall return to its previous firmware.	H-7 control 1; Scenario-450	TEST
REQ-014	New firmware that hangs during its trial shall be given up, and the controller shall return to its previous firmware.	H-8 control 4; Scenario-430	TEST
REQ-015	After [**TBD-3**] failed starts of new firmware, the controller shall return to its previous firmware.	H-1 control 3; Scenario-440	TEST
REQ-016	A power loss during the trial shall count as a failed start.	H-1 control 7; Scenario-605	TEST
REQ-017	The count of failed starts shall survive a restart and a power loss.	H-1 control 3	TEST
REQ-018	When the previous firmware also fails, the controller shall run its last-resort firmware.	Scenario-460	TEST
REQ-019	Failed firmware shall not be started again unless the operator orders it.	H-1 control 8	TEST
REQ-020	The controller shall not confirm firmware that has not been judged good.	Must not work: confirm firmware not judged good	TEST
REQ-021	The controller shall switch to new firmware at the moment set in [**TBD-4**].	Scenario-100; Capability-4	TEST
REQ-022	The controller firmware shall restart on its own when it stops responding, and resume operation.	H-8 control 1; Scenario-620	TEST
REQ-023	A restart of the controller shall complete within [**TBD-5**], shorter than the time to the next failure.	H-8 cause d; SWE-134 j	TEST
REQ-024	The controller firmware shall keep working while the whole onboard computer restarts.	H-8 control 5; Scenario-535	TEST
REQ-025	The controller firmware shall keep working on its current firmware when both data-link connections fail.	Scenario-510	TEST
REQ-026	The controller firmware shall keep working correctly however long no command arrives.	H-8 cause e	TEST
REQ-027	The controller firmware shall check that every command is allowed in the current mode, and refuse it with the reason if not.	H-6 control 1; Scenario-310; SWE-134 e	TEST
REQ-028	The controller firmware shall check that every command has possible values, and refuse it with the reason if not.	H-6 control 1; Scenario-320; SWE-134 g	TEST
REQ-029	The controller firmware shall run a dangerous command only after a valid arming command, and refuse it with the reason otherwise.	H-6 control 2; Scenario-330; SWE-134 d; SWE-134 h	TEST

ID	Requirement	Source	Verified by
REQ-030	An arming command shall expire when the dangerous command does not follow within [**TBD-6**]; a later dangerous command shall be refused with the reason.	H-6 control 2; Scenario-335	TEST
REQ-031	The controller firmware shall run a command sent again only once, and answer “already done”.	H-6 control 3; Scenario-350	TEST
REQ-032	The controller firmware shall check a held command again just before running it.	H-6 cause i; Shared cause: radiation	TEST
REQ-033	The controller firmware shall send two answers to every command: received, then done with the result.	H-11 control 1; Scenario-120; Capability-2	TEST
REQ-034	The controller firmware shall answer “done” only after the command has run.	H-9 cause d	TEST
REQ-035	The controller firmware shall answer every refused command with the reason.	H-11 control 2; Scenario-340	TEST
REQ-036	The controller firmware shall log every command it receives and every answer it sends.	H-6 control 5; Capability-2	INSPECTION
REQ-037	The controller firmware shall send regular health reports at [**TBD-7**] and an event report as soon as an event happens.	Scenario-110; Capability-1; O-5	TEST
REQ-038	The controller firmware shall detect faults between contacts, respond without the ground, and report each fault in an event report.	Capability-3; O-4; H-10 cause b; H-13	TEST
REQ-039	The controller firmware shall test itself at every start and at [**TBD-8**], and report a failed test.	H-10 cause a	TEST
REQ-040	After a fall-back, the next reports shall state which firmware runs.	H-9 control 4; Scenario-430	TEST
REQ-041	The controller firmware shall detect a change in its own memory and return to a known safe state.	SWE-134 f	TEST
REQ-042	The controller firmware shall never send on the data link in a way that stops the onboard computer reaching it.	H-4 cause c	TEST
REQ-113	The controller shall return to its previous firmware when the operator orders it.	H-7 control 5	TEST
REQ-117	In FALLBACK, the controller shall keep running the firmware it fell back to until the operator sends new firmware or a command that changes its firmware or mode.	Mode table; Scenario-460	TEST
REQ-119	The bootloader shall share no code with the main firmware or the last-resort firmware.	Architecture 5.1	INSPECTION
REQ-125	No controller firmware shall be able to disable the debug port or the factory loader.	Architecture 5.2	INSPECTION
REQ-130	The controller firmware shall check firmware signatures with code different from the onboard computer software’s.	Architecture 5.4	INSPECTION
REQ-131	The controller firmware shall keep its own clock, from a time source separate from the onboard computer’s.	Architecture 5.5	TEST
REQ-132	The controller’s clock shall be set only by time values sent from the ground, never from the onboard computer’s clock.	Architecture 5.5	TEST
REQ-133	The controller firmware shall refuse a command past its expiry, by its own clock.	Architecture 5.5	TEST
REQ-134	The controller firmware shall run a scheduled command only when its own clock agrees with the scheduled time within [**TBD-13**].	Architecture 5.5	TEST
REQ-139	During its trial, new firmware shall test itself and report the result: its self-test, its answers on the data link, and the receipt of a test image with a correct check value.	H-3 control 2; Scenario-100; Capability-4	TEST

ID	Requirement	Source	Verified by
REQ-140	New firmware that fails a trial test shall be given up at once, and the controller shall return to its previous firmware.	Scenario-445	TEST

5.0 Onboard Computer Software Requirements

ID	Requirement	Source	Verified by
REQ-043	The onboard computer software shall refuse firmware not released by the operator.	H-2 control 1; Scenario-410; O-2	TEST
REQ-044	The onboard computer software shall refuse firmware damaged in transfer.	H-7 control 2; Scenario-400	TEST
REQ-045	The onboard computer software shall refuse firmware older than the minimum version.	H-7 control 2; Scenario-420	TEST
REQ-046	The onboard computer software shall refuse firmware built for a different controller.	H-7 control 2; Scenario-360	TEST
REQ-047	The onboard computer software shall check firmware after its last transfer, just before delivering it.	H-7 control 2	TEST
REQ-048	The onboard computer software shall deliver firmware to the controller while the controller keeps working.	Scenario-100	TEST
REQ-049	The onboard computer software shall resume an interrupted upload at the next contact, from where it stopped.	Scenario-200; H-14	TEST
REQ-050	The onboard computer software shall not deliver new firmware during a trial, and shall tell the ground why.	Scenario-425; Mode table	TEST
REQ-051	The onboard computer software shall stop an update when the operator cancels it, leaving working firmware running.	Scenario-470	TEST
REQ-052	The onboard computer software shall free its firmware storage when an update ends or is cancelled.	H-3 cause d; H-14	TEST
REQ-053	The onboard computer software shall let the operator correct the stored minimum version and controller identity.	H-3 cause c	TEST
REQ-054	The onboard computer software shall restore the last-resort firmware through the reprogramming link when no firmware can run.	H-1 control 5; Scenario-480; O-6	TEST
REQ-055	The onboard computer software shall restore the last-resort firmware when the operator orders it, also while firmware runs.	H-2 control 3; H-3 control 4	DEMONSTRATION
REQ-056	The restore shall not depend on any software of the controller.	H-1 control 5; Shared cause: image code and data	ANALYSIS
REQ-057	The restore shall clear the data the controller keeps over restarts.	Shared cause: image code and data	TEST
REQ-058	The onboard computer software shall be able to load new firmware from the ground through the reprogramming link.	Shared cause: image code and data; H-8 cause g	TEST
REQ-059	The onboard computer software shall check its copy of the last-resort firmware and the reprogramming link between uses, and report any fault.	H-1 control 6	TEST
REQ-060	The onboard computer software shall tell the ground at the next contact that a restore took place, and its result.	Scenario-480	TEST

ID	Requirement	Source	Verified by
REQ-061	The onboard computer software shall notice when the controller goes silent, try to bring it back, and record what it did.	H-8 control 2; Scenario-520	TEST
REQ-062	The onboard computer software shall restart a crashed program and resume any update in progress.	H-8 control 3; Scenario-530; H-14	TEST
REQ-063	After a restart of the whole onboard computer, its software shall resume any update in progress.	Scenario-535; H-14	TEST
REQ-064	The onboard computer software shall check that every command comes from the operator, and refuse and log any other.	H-5 control 1; Scenario-300; O-2	TEST
REQ-065	The onboard computer software shall refuse an old command sent again.	H-6 control 1; Scenario-300	TEST
REQ-066	The onboard computer software shall refuse a command past its expiry, with the reason.	H-6 control 1; Scenario-340	TEST
REQ-067	The onboard computer software shall accept commands only from the radio link from the ground.	H-5 control 1	TEST
REQ-068	The onboard computer software shall hold scheduled commands and pass each to the controller at its run time.	Scenario-130; Capability-2	TEST
REQ-069	A scheduled command shall survive a restart and a power loss, and run once only.	H-11 control 3; H-6 cause h	TEST
REQ-070	The onboard computer software shall time-tag every report.	H-9 control 1; Scenario-110	TEST
REQ-071	The onboard computer software shall number every report, so that its order does not depend on the onboard clock.	H-9 cause a; Shared cause: wrong onboard clock	TEST
REQ-072	The onboard computer software shall compare the onboard clock with ground time at every contact, and report a difference above [**TBD-9**].	H-6 cause h; H-9 cause a; Shared cause: wrong onboard clock	TEST
REQ-073	The onboard computer software shall accept a correction of the onboard clock that does not depend on the onboard clock.	H-4 control 4	TEST
REQ-074	The onboard computer software shall store every report between contacts and send all of them at the next contact.	Scenario-110; Capability-1	TEST
REQ-075	When report storage is full, the onboard computer software shall drop the oldest health reports first and keep every event report.	H-10 control 1; Scenario-210; H-15	TEST
REQ-076	The onboard computer software shall tell the ground how many health reports were dropped, and the time span they covered.	H-10 control 2; Scenario-210; O-5; H-15	TEST
REQ-077	Stored reports shall survive a power loss and a restart, including a restart of the whole onboard computer.	H-10 control 3; Scenario-535; H-15	TEST
REQ-078	At a contact, the onboard computer software shall send event reports before health reports.	Scenario-210; H-16	TEST
REQ-079	The onboard computer software shall check every report from the controller for damage when it arrives.	H-9 control 2	TEST
REQ-080	The onboard computer software shall reach the controller over two separate data-link connections, either one being enough.	H-4 control 1; Scenario-500; O-3	TEST
REQ-081	The onboard computer software shall report the loss of a data-link connection.	H-4 control 2; Scenario-500; Scenario-510	TEST
REQ-082	The onboard computer software shall use a failed connection again once it works, and report it.	H-4 control 5; Scenario-505; H-13	TEST
REQ-083	The onboard computer software shall keep the reprogramming link usable when both data-link connections fail.	H-4 control 3; Scenario-510	TEST

ID	Requirement	Source	Verified by
REQ-084	The onboard computer software shall accept a new means of proving origin only in a command proven with the current one.	Scenario-370	TEST
REQ-085	After a replacement of the means of proving origin, only the new means shall be accepted.	H-2 control 4; H-5 control 3; Scenario-370	TEST
REQ-086	A replacement of the means of proving origin interrupted at any point shall leave the current means working.	H-4 cause g	TEST
REQ-087	The onboard computer software shall send the ground a log of every command acted on, so that a command the operator did not send can be noticed.	Shared cause: one proof of origin	TEST
REQ-112	The onboard computer software shall pass a dangerous command to the controller only after a valid arming command that has not expired, and refuse it with the reason otherwise.	H-6 control 6; Scenario-330; Scenario-335	TEST
REQ-118	The restore shall write to the controller only images, the bootloader and the last-resort firmware included, that have passed the same checks as an update: origin, damage, minimum version and target controller.	H-2 control 1; H-7 control 2	TEST
REQ-120	The restore shall put the last-resort firmware in the bootloader's place, so that it runs first.	Architecture 5.1	TEST
REQ-121	The onboard computer software shall put back the bootloader and the main firmware through the reprogramming link when the operator orders it.	Architecture 5.1	TEST
REQ-122	The reprogramming link shall have two connections, independent of each other: the controller's debug port and its factory loader, each able to load the controller's memory whatever that memory holds.	Architecture 5.2	TEST
REQ-123	The onboard computer software shall drive the debug port and the factory loader with separate drivers that share no code.	Architecture 5.2	INSPECTION
REQ-124	The onboard computer software shall be able to restart the controller.	Architecture 5.2	TEST
REQ-141	The onboard computer software shall drive the reprogramming link only with code developed and verified to these requirements; no general-purpose debug tool shall be part of the flight software.	Architecture 5.2	INSPECTION
REQ-126	The restore program shall share no code or stored data with the update manager.	Architecture 5.3	INSPECTION
REQ-129	The onboard computer software shall act on a command or firmware image only when it decrypts and both signatures are valid.	Architecture 5.4	TEST

6.0 Ground Software Requirements

ID	Requirement	Source	Verified by
REQ-088	The ground software shall prove that every command and firmware release comes from the operator.	Scenario-300; O-2	TEST
REQ-089	The ground software shall check every command before it is sent.	H-6 control 4; Scenario-120	TEST

ID	Requirement	Source	Verified by
REQ-090	The ground software shall check every firmware release before it is sent: built for this controller, not older than the minimum version, and the build that was tested.	H-7 control 4; Scenario-100	TEST
REQ-091	The ground software shall let only the operator use the operator interface.	H-5 control 2	DEMONSTRATION
REQ-092	The ground software shall keep every report received.	H-9 control 3; H-10 control 4; O-5	INSPECTION
REQ-093	The ground software shall check every report for damage when it arrives.	H-9 control 2	TEST
REQ-094	The ground software shall show the operator any command whose answers have not arrived.	H-11 control 1	TEST
REQ-095	The ground software shall show the operator which firmware runs on the controller.	H-9 control 4	TEST
REQ-096	The ground software shall keep a recoverable copy of the means of proving origin.	H-4 cause e	INSPECTION
REQ-097	The ground software shall let the operator plan commands, uploads and report retrieval for coming contacts [**TBD-10**].	Capability-2; Capability-4; O-1; H-16	DEMONSTRATION
REQ-114	The ground software shall check every signed firmware image with the checks the onboard computer software makes, before it is sent.	H-14; Capability-4	TEST
REQ-115	The ground software shall let the operator send a planned sequence of commands with one action.	Capability-2	TEST
REQ-116	In a sequence, the ground software shall send each command only after the previous one is answered done, and stop at the first command refused or not answered.	H-6 control 7	TEST
REQ-128	Every command and firmware image shall carry two signatures, made with two keys kept apart from each other and from the encryption key.	Architecture 5.4	INSPECTION

7.0 Requirements for All Software Items

These apply to all three items. Most are NASA's rules for safety-critical software [1, Secs. 3.7.3–3.7.5, pp. 26–27; Secs. 4.4.3–4.4.4, p. 34].

ID	Requirement	Source	Verified by
REQ-098	A catastrophic hazard shall need at least three failures together to occur, and a critical hazard at least two.	Fault tolerance rule	ANALYSIS
REQ-099	The two data-link connections shall share no software, so that one defect cannot cut both.	H-4 cause b; Shared cause: same data-link software	INSPECTION
REQ-100	The ground and the onboard software shall use one shared list of commands and refusal reasons.	H-17	INSPECTION
REQ-101	The operating software shall contain no fault-injection function; test functions shall run only on the bench.	H-12 control 1	INSPECTION
REQ-102	Every software function that ends shall leave the System in a known safe state.	SWE-134 c	TEST
REQ-103	No single software event or action shall start an identified hazard.	SWE-134 i	ANALYSIS
REQ-104	The software shall respond to each off-nominal condition within the time needed to prevent a hazard [**TBD-11**].	SWE-134 j	ANALYSIS
REQ-105	The software shall handle every error it detects.	SWE-134 k	TEST
REQ-106	The software shall be able to place the System into a safe state.	SWE-134 l	TEST

ID	Requirement	Source	Verified by
REQ-107	The software shall check the integrity of every input and output on each link.	SWE-134 g	TEST
REQ-108	The safety-critical software shall reach 100 % modified condition/decision coverage in test.	SWE-219	ANALYSIS
REQ-109	Each safety-critical software component shall have a cyclomatic complexity of 15 or lower.	SWE-220	INSPECTION
REQ-110	All software shall follow the coding standard [**TBD-12**].	SWE-061	INSPECTION
REQ-111	All software shall be analysed with static analysis tools for defects, security, coverage and complexity.	SWE-135	INSPECTION
REQ-127	Every command and firmware image shall be encrypted from the ground software to the onboard computer software.	Architecture 5.4	TEST
REQ-135	Every report shall carry the time of both clocks.	Architecture 5.5	TEST
REQ-136	Every held command and the kept data shall be stored with a check value and checked just before use.	Architecture 5.6	TEST
REQ-137	The arm state shall be stored so that no single bit error makes it armed.	Architecture 5.6	TEST
REQ-138	The two data-link connections shall be of different kinds.	Architecture 5.7	INSPECTION

A Coverage

Every item the earlier documents ask of the software, and how it is covered: by the requirements listed, by something outside this project (OUTSIDE), or as an accepted risk (ACCEPTED). No item is left uncovered.

A.1 Hazard Controls

Item	Requirements	Item	Requirements	Item	Requirements
H-1 control 1	REQ-007	H-1 control 2	REQ-005	H-1 control 3	REQ-015, REQ-017
H-1 control 4	REQ-008	H-1 control 5	REQ-054, REQ-056	H-1 control 6	REQ-059
H-1 control 7	REQ-016	H-1 control 8	REQ-019	H-2 control 1	REQ-043, REQ-118
H-2 control 2	REQ-006	H-2 control 3	REQ-055	H-2 control 4	REQ-085
H-3 control 1	REQ-011	H-3 control 2	REQ-012, REQ-139	H-3 control 3	REQ-010
H-3 control 4	REQ-055	H-3 control 5	OUTSIDE	H-4 control 1	REQ-080
H-4 control 2	REQ-081	H-4 control 3	REQ-083	H-4 control 4	REQ-073
H-4 control 5	REQ-082	H-5 control 1	REQ-064, REQ-067	H-5 control 2	REQ-091
H-5 control 3	REQ-085	H-6 control 1	REQ-027, REQ-028, REQ-065, REQ-066	H-6 control 2	REQ-029, REQ-030
H-6 control 3	REQ-031	H-6 control 4	REQ-089	H-6 control 5	REQ-036
H-6 control 6	REQ-112	H-6 control 7	REQ-116	H-7 control 1	REQ-013
H-7 control 2	REQ-044, REQ-045, REQ-046, REQ-047, REQ-118	H-7 control 3	REQ-005	H-7 control 4	REQ-090
H-7 control 5	REQ-113	H-8 control 1	REQ-022	H-8 control 2	REQ-061
H-8 control 3	REQ-062	H-8 control 4	REQ-014	H-8 control 5	REQ-024

Item	Requirements	Item	Requirements	Item	Requirements
H-9 control 1	REQ-070	H-9 control 2	REQ-079, REQ-093	H-9 control 3	REQ-092
H-9 control 4	REQ-040, REQ-095	H-10 control 1	REQ-075	H-10 control 2	REQ-076
H-10 control 3	REQ-077	H-10 control 4	REQ-092	H-11 control 1	REQ-033, REQ-094
H-11 control 2	REQ-035	H-11 control 3	REQ-069	H-12 control 1	REQ-101

A.2 Hazard Causes

Causes the hazard analysis left without a control, and causes a requirement answers directly.

Item	Requirements
H-1 cause g	REQ-009
H-3 cause c	REQ-053
H-3 cause d	REQ-052
H-4 cause b	REQ-099
H-4 cause c	REQ-042
H-4 cause e	REQ-096
H-4 cause g	REQ-086
H-4 cause h	ACCEPTED
H-6 cause h	REQ-069, REQ-072
H-6 cause i	REQ-032
H-8 cause d	REQ-023
H-8 cause e	REQ-026
H-8 cause g	REQ-009, REQ-058
H-9 cause a	REQ-071, REQ-072
H-9 cause d	REQ-034
H-10 cause a	REQ-039
H-10 cause b	REQ-038

A.3 Must-Work and Must-Not-Work Functions

Item	Requirements
Return to the previous firmware when new firmware fails	REQ-013, REQ-014
Run the last-resort firmware when the previous one fails	REQ-018
Restore through the reprogramming link	REQ-054
Check every firmware image before it runs	REQ-005
Refuse firmware and commands not from the operator	REQ-064
Accept the clock correction	REQ-073
Bring a silent controller back	REQ-061
Detect faults, and keep and send down every event report	REQ-038, REQ-074
Answer every command	REQ-033
Tell the ground which firmware runs	REQ-040
Replace a disclosed means of proving origin	REQ-085
Write over the previous or last-resort firmware	REQ-007
Start failed firmware again without an operator order	REQ-019
Run firmware not from the operator	REQ-006, REQ-043
Run a command not from the operator	REQ-064
Run a dangerous command without a valid arm	REQ-029, REQ-112
Run a command twice, past its expiry, or at the wrong time	REQ-031, REQ-066
Confirm firmware not judged good	REQ-020
Inject faults	REQ-101
Accept a new means of proving origin not proven with the current one	REQ-084

A.4 ConOps Scenarios

Item	Requirements	Item	Requirements	Item	Requirements
Scenario-100	REQ-011, REQ-012, REQ-021, REQ-048, REQ-090, REQ-139	Scenario-110	REQ-037, REQ-070, REQ-074	Scenario-120	REQ-033, REQ-089
Scenario-130	REQ-068	Scenario-200	REQ-049	Scenario-210	REQ-075, REQ-076, REQ-078
Scenario-300	REQ-064, REQ-065, REQ-088	Scenario-310	REQ-027	Scenario-320	REQ-028
Scenario-330	REQ-029, REQ-112	Scenario-335	REQ-030, REQ-112	Scenario-340	REQ-035, REQ-066
Scenario-350	REQ-031	Scenario-360	REQ-046	Scenario-370	REQ-084, REQ-085
Scenario-400	REQ-044	Scenario-410	REQ-043	Scenario-420	REQ-045
Scenario-425	REQ-002, REQ-050	Scenario-430	REQ-014, REQ-040	Scenario-440	REQ-015
Scenario-445	REQ-140	Scenario-450	REQ-013	Scenario-460	REQ-008, REQ-018, REQ-117
Scenario-470	REQ-051	Scenario-480	REQ-054, REQ-060	Scenario-500	REQ-080, REQ-081
Scenario-505	REQ-082	Scenario-510	REQ-025, REQ-081, REQ-083	Scenario-520	REQ-061
Scenario-530	REQ-062	Scenario-535	REQ-024, REQ-063, REQ-077	Scenario-600	REQ-007
Scenario-605	REQ-016	Scenario-610	REQ-005	Scenario-620	REQ-022

A.5 ConOps Capabilities, Objectives and Modes

Item	Requirements	Item	Requirements	Item	Requirements
Capability-1	REQ-037, REQ-074	Capability-2	REQ-033, REQ-036, REQ-068, REQ-097, REQ-115	Capability-3	REQ-038
Capability-4	REQ-011, REQ-021, REQ-097, REQ-114	Capability-5	OUTSIDE	O-1	REQ-097
O-2	REQ-043, REQ-064, REQ-088	O-3	REQ-080	O-4	REQ-038
O-5	REQ-037, REQ-076, REQ-092	O-6	REQ-005, REQ-007, REQ-054	Mode table	REQ-001, REQ-002, REQ-003, REQ-010, REQ-050, REQ-117

A.6 NASA Rules for Safety-Critical Software

Item	Requirements	Item	Requirements	Item	Requirements
SWE-134 a	REQ-004	SWE-134 b	REQ-001, REQ-003	SWE-134 c	REQ-102
SWE-134 d	REQ-029	SWE-134 e	REQ-027	SWE-134 f	REQ-041
SWE-134 g	REQ-028, REQ-107	SWE-134 h	REQ-029	SWE-134 i	REQ-103
SWE-134 j	REQ-023, REQ-104	SWE-134 k	REQ-105	SWE-134 l	REQ-106
SWE-219	REQ-108	SWE-220	REQ-109	SWE-061	REQ-110
SWE-135	REQ-111				

A.7 Hazard Analysis Findings and Rules

Item	Requirements
Shared cause: image code and data	REQ-009, REQ-056, REQ-057, REQ-058
Shared cause: one proof of origin	REQ-087
Shared cause: onboard computer software	OUTSIDE
Shared cause: one onboard computer	ACCEPTED
Shared cause: same data-link software	REQ-099
Shared cause: wrong onboard clock	REQ-071, REQ-072
Shared cause: radiation	REQ-032
Fault tolerance rule	REQ-098

A.8 Moderate and Negligible Hazards

These hazards have no hazard report [5, Sec. 6.0]; each is listed with the requirements that address it.

Item	Requirements
H-13	REQ-038, REQ-082
H-14	REQ-049, REQ-052, REQ-062, REQ-063, REQ-114
H-15	REQ-075, REQ-076, REQ-077
H-16	REQ-078, REQ-097
H-17	REQ-100

A.9 Architecture Decisions

The design decisions of the architecture [7, Sec. 5.0], each with the requirements it adds.

Item	Requirements
Architecture 5.1	REQ-119, REQ-120, REQ-121
Architecture 5.2	REQ-122, REQ-123, REQ-124, REQ-125, REQ-141
Architecture 5.3	REQ-126
Architecture 5.4	REQ-127, REQ-128, REQ-129, REQ-130
Architecture 5.5	REQ-131, REQ-132, REQ-133, REQ-134, REQ-135
Architecture 5.6	REQ-136, REQ-137
Architecture 5.7	REQ-138

B Open Items

The values below are not yet known.

TBD-1 Project name and document reference.
TBD-2 Trial time limit (ConOps TBD-4).
TBD-3 Failed starts before going back (ConOps TBD-5).
TBD-4 Switch on arrival, or only on operator command (ConOps TBD-2).
TBD-5 Longest time a controller restart may take.
TBD-6 Time after which an arming command expires.
TBD-7 Rate of regular health reports.
TBD-8 Interval between controller self-tests.
TBD-9 Largest allowed difference between the onboard clock and ground time.
TBD-10 Contact window length and gap (ConOps TBD-3).
TBD-11 Response time for each off-nominal condition.
TBD-12 The coding standard.
TBD-13 Largest allowed difference between the controller's clock and the time of a scheduled command.

The author accepted the allocation of functions in Section 3.1 on 11 October 2026.